

EXPERIMENT 2

AIM

To add the constraints like **PRIMARY KEY**, **FOREIGN KEY**, **UNIQUE KEY**, and **CHECK** using Data Definition Language (DDL) commands.

Description

PRIMARY KEY

- The **PRIMARY KEY** constraint uniquely identifies each record in a table.
- Primary key values must be **UNIQUE** and **NOT NULL**.
- A table can have only **one PRIMARY KEY**.

FOREIGN KEY

- A **FOREIGN KEY** is used to establish a relationship between two tables.
- It refers to the **PRIMARY KEY** of another table.
- The table containing the foreign key is called the **child table**, and the referenced table is called the **parent table**.

UNIQUE Constraint

- The **UNIQUE** constraint ensures that all values in a column are different.
- A table can have multiple **UNIQUE** constraints.
- A **PRIMARY KEY** automatically has a **UNIQUE** constraint.

CHECK Constraint

- The **CHECK** constraint restricts the values that can be stored in a column.
- It ensures that only valid data is inserted into the table.

Syntax

PRIMARY KEY

ALTER TABLE table_name

ADD PRIMARY KEY(column_name);

FOREIGN KEY

ALTER TABLE table_name

ADD CONSTRAINT constraint_name

FOREIGN KEY(column_name)

REFERENCES parent_table(column_name);

UNIQUE

ALTER TABLE table_name

ADD CONSTRAINT constraint_name

UNIQUE(column_name);

CHECK

ALTER TABLE table_name

ADD CONSTRAINT constraint_name

CHECK(condition);

Questions

1) Alter the table HOSTEL_ROOMS2 with the following structure.

S.No	Column Name	Constraint
1	RoomID	PRIMARY KEY

2) Alter the table HOSTEL_STUDENT2 with the following structure.

S.No	Column Name	Constraint
1	StudentID	PRIMARY KEY
2	Gender	CHECK ('M' or 'F')

3) Alter the table HOSTEL_WARDEN2 with the following structure.

S.No	Column Name	Constraint
1	WardenID	PRIMARY KEY

4) Alter the table HOSTEL_BLOCK2 with the following structure.

S.No	Column Name	Constraint
1	BlockID	PRIMARY KEY
2	FloorCount	CHECK (Room Number between 1 and 50)

5) Alter the table HOSTEL_ROOMS2 with the following structure.

S.No	Column Name	Constraint
1	MobileNo	UNIQUE KEY

6) After the HOSTEL_ROOMS2, HOSTEL_STUDENT2, and HOSTEL_BLOCK2 tables are successfully created, add FOREIGN KEY constraints to the HOSTEL_BLOCK2 table.

S.No	Child Table	Foreign Key	Parent Table	Primary Key
1	HOSTEL_BLOCK2	RoomID	HOSTEL_ROOMS2	RoomID
2	HOSTEL_BLOCK2	StudentID	HOSTEL_STUDENT2	StudentID

Question 1

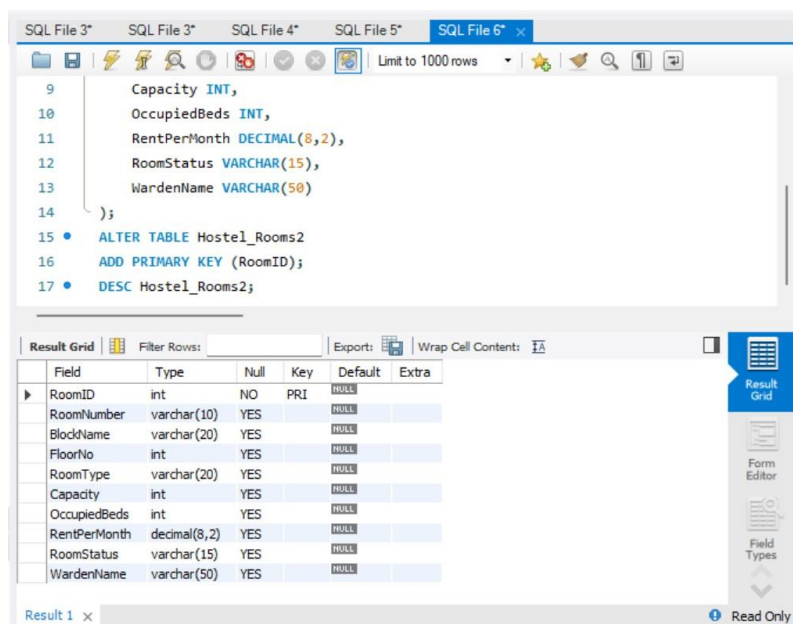
Alter the table HOSTEL_ROOMS2 by adding RoomID as the PRIMARY KEY.

```
ALTER TABLE HOSTEL_ROOMS2
```

```
ADD PRIMARY KEY (RoomID);
```

Verification

```
DESC HOSTEL_ROOMS2;
```



Question 2

Alter the table HOSTEL_STUDENT2 by adding StudentID as the PRIMARY KEY and Gender as a CHECK constraint ('M' or 'F').

Step 1: Add Primary Key

```
ALTER TABLE HOSTEL_STUDENT2
```

```
ADD PRIMARY KEY (StudentID);
```

Step 2: Add CHECK Constraint

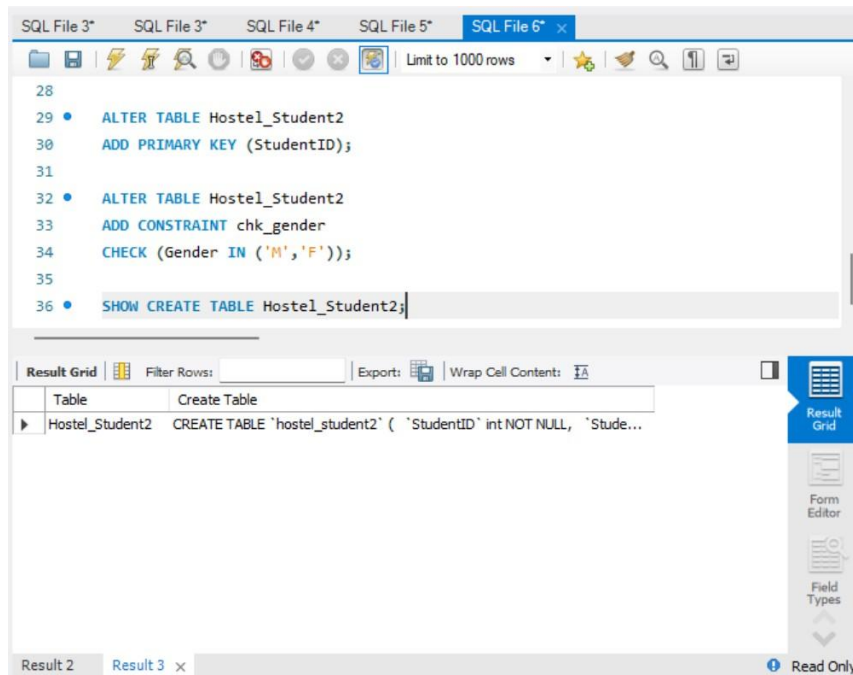
```
ALTER TABLE HOSTEL_STUDENT2
```

```
ADD CONSTRAINT chk_gender
```

```
CHECK (Gender IN ('M','F'));
```

Verification

```
SHOW CREATE TABLE HOSTEL_STUDENT2;
```



Question 3

Alter the table HOSTEL_WARDEN2 by adding WardenID as the PRIMARY KEY.

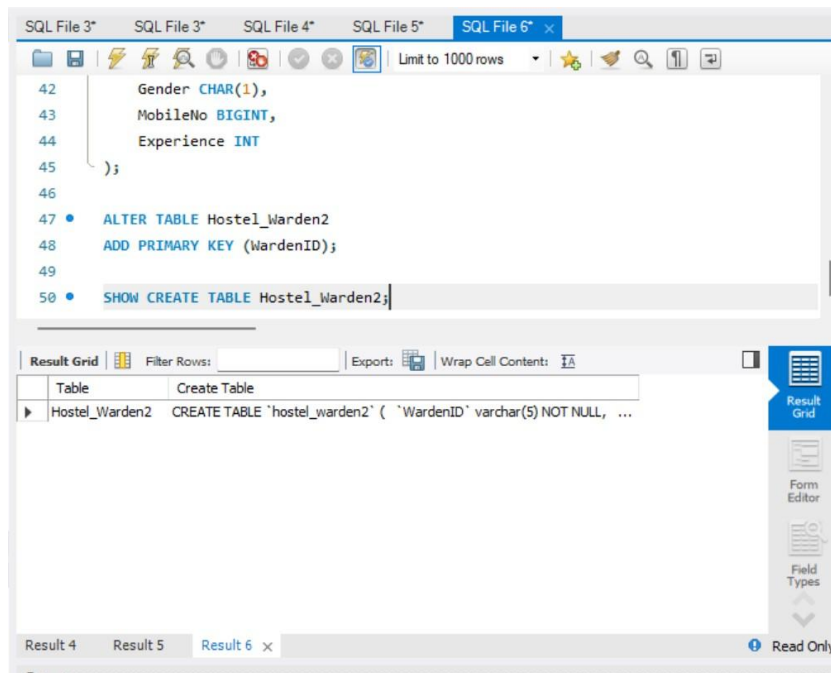
```
ALTER TABLE HOSTEL_WARDEN2
```

```
ADD PRIMARY KEY (WardenID);
```

Verification

```
DESC HOSTEL_WARDEN2;
```

```
SHOW CREATE TABLE HOSTEL_WARDEN2;
```



Question 4

Alter the table HOSTEL_BLOCK2 by adding BlockID as the PRIMARY KEY and a CHECK constraint for FloorCount (between 1 and 50).

Step 1: Add Primary Key

```
ALTER TABLE HOSTEL_BLOCK2
```

```
ADD PRIMARY KEY (BlockID);
```

2: Add CHECK Constraint

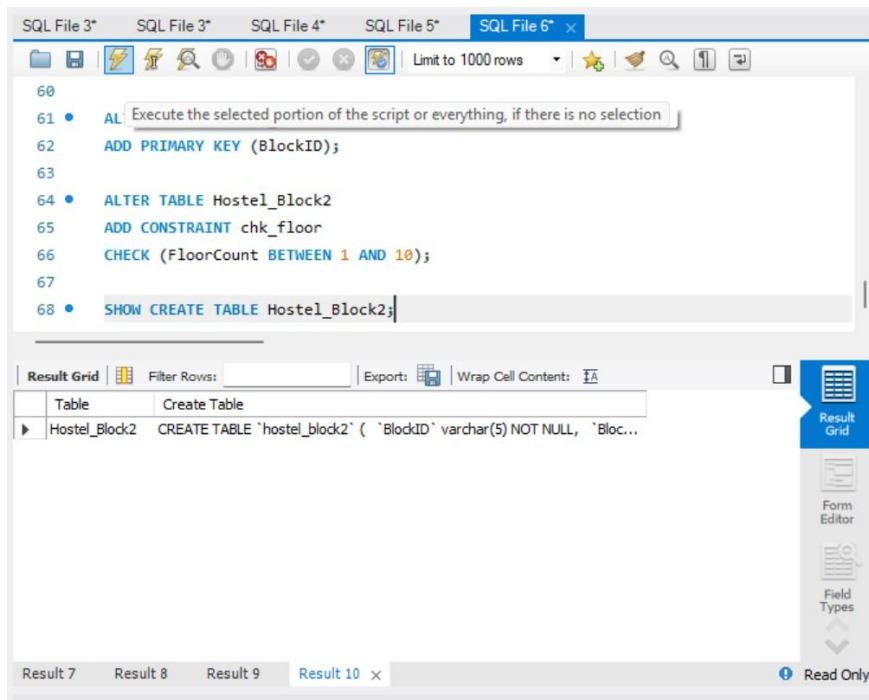
```
ALTER TABLE HOSTEL_BLOCK2
```

```
ADD CONSTRAINT chk_room
```

```
CHECK (CAST(FloorCount AS UNSIGNED) BETWEEN 1 AND 50);
```

Verification

```
SHOW CREATE TABLE HOSTEL_BLOCK2;
```



Question 5

Alter the table HOSTEL_ROOMS2 by adding a UNIQUE constraint to MobileNo.

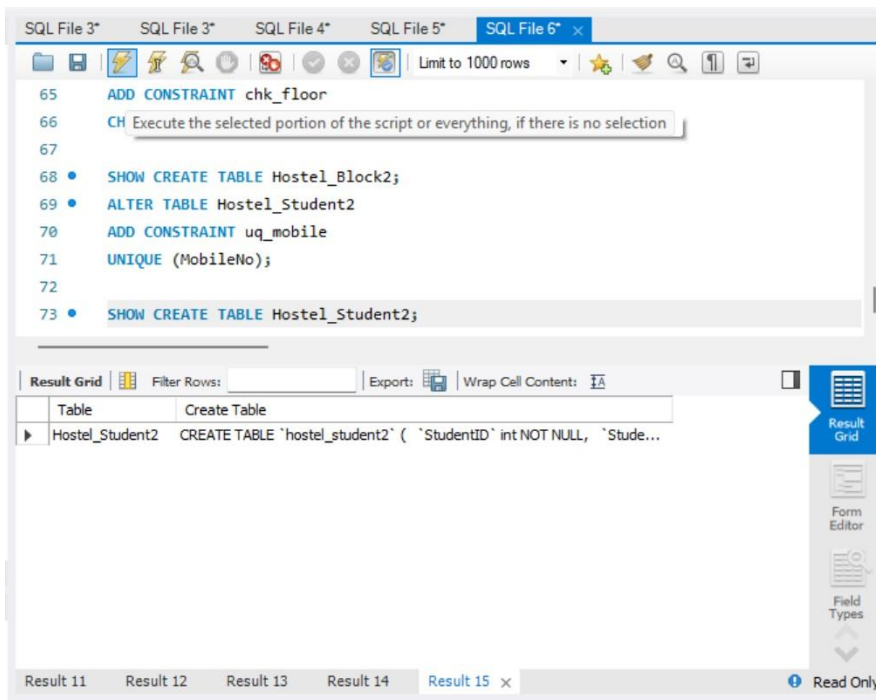
```
ALTER TABLE HOSTEL_ROOMS2
```

```
ADD CONSTRAINT uq_mobile
```

```
UNIQUE (MobileNo);
```

Verification

```
SHOW CREATE TABLE HOSTEL_ROOMS2;
```



Question 6

Add FOREIGN KEY constraints to the HOSTEL_BLOCK2 table.

Step 1: Add Foreign Key for RoomID

```
ALTER TABLE HOSTEL_BLOCK2
```

```
ADD CONSTRAINT fk_patient
```

```
FOREIGN KEY (RoomID)
```

```
REFERENCES HOSTEL_ROOMS2(RoomID);
```

Step 2: Add Foreign Key for StudentID

```
ALTER TABLE HOSTEL_BLOCK2
```

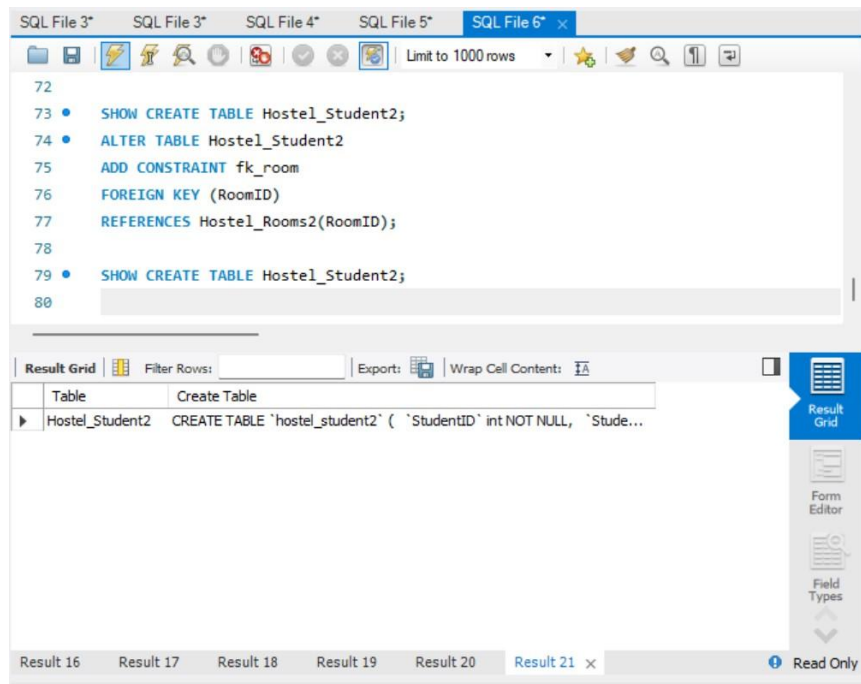
```
ADD CONSTRAINT fk_doctor
```

```
FOREIGN KEY (StudentID)
```

```
REFERENCES HOSTEL_STUDENT2(StudentID);
```

Verification

```
SHOW CREATE TABLE HOSTEL_BLOCK2;
```



Result

The PRIMARY KEY, FOREIGN KEY, UNIQUE, and CHECK constraints were successfully added to the HOSTEL_ROOMS2, HOSTEL_STUDENT2, HOSTEL_WARDEN2, and HOSTEL_BLOCK2 tables using DDL commands. The constraints were verified successfully in MySQL Workbench.